

PE-EC603A — Introduction to MEMS

Semester VI • 3L:0T:0P • 3 Credits

Module 4 Exam Notes: Bulk Micromachining & Mechanics

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1. Bulk Micromachining & Wafer Bonding

Bulk micromachining fabricates micro-structures directly inside the substrate by selectively etching away deep portions of the silicon wafer.

1.1 Wet Isotropic vs. Anisotropic Etching

- **Isotropic Etching (HNA):**

- Uses a mixture of Hydrofluoric acid (HF), Nitric acid (HNO_3), and Acetic acid (CH_3COOH) — known as **HNA**.
- **Mechanism:** Nitric acid oxidizes the silicon surface to form SiO_2 , Hydrofluoric acid dissolves this oxide, and Acetic acid acts as a diluent to control reaction rate.
- Etches equally in all directions, leading to rounded cavities and mask undercutting.

- **Anisotropic Etching (KOH / TMAH):**

- Uses Potassium Hydroxide (**KOH**) or Tetramethylammonium Hydroxide (**TMAH**).
- Etches along crystallographic planes. The etch rate is extremely slow on the $\{111\}$ plane compared to $\{100\}$ (selectivity up to 400 : 1 for KOH).
- **Reason for Slow $\{111\}$ Etch:** The $\{111\}$ plane has the highest atomic packing density, and surface atoms on $\{111\}$ present only a single dangling bond to the chemical solution (compared to two bonds on $\{100\}$), making it highly resistant to chemical attack.
- Results in precise V-grooves and pyramidal cavities with sidewalls sloped at an angle of 54.74° relative to the $\{100\}$ surface.

1.2 Wafer Bonding Techniques

Wafer bonding joins two substrates to form complex 3D structures or sealed packages:

1. **Direct / Fusion Bonding:** Joins silicon to silicon. Silicon surfaces are polished, hydrated (creating silanol $Si - OH$ groups), pressed together, and annealed at high temperature ($> 1000^\circ C$). The silanol groups polymerize to form strong covalent covalent bonds ($Si - O - Si$).
2. **Anodic Bonding:** Joins silicon to sodium-rich glass (e.g., Pyrex 7740). The substrates are heated to $300^\circ C - 450^\circ C$, and a high DC voltage (400 V–1000 V) is applied with glass as cathode. Sodium ions (Na^+) migrate away from the interface, creating a depletion region and strong electrostatic pull that initiates chemical bonding.
3. **Eutectic Bonding:** Joins silicon to metal (e.g., Gold). They are heated to the eutectic temperature ($363^\circ C$ for $Au-Si$), forming a liquid alloy phase that solidifies to create a high-strength hermetic seal.

2. Solid Mechanics at Microscale

Understanding mechanical deformation, stress, and strain is essential for designing MEMS suspensions.

2.1 Stress, Strain, and Elastic Constants

- **Stress (σ):** Force per unit area ($\sigma = F/A$ in N/m^2 or Pa).
- **Strain (ϵ):** Fractional elongation ($\epsilon = \Delta L/L$, dimensionless).
- **Hooke's Law:** States that within the elastic limit, stress (σ) is directly proportional to strain (ϵ).
 - **1D Formulation:** $\sigma = E\epsilon$ (where E is Young's Modulus, a measure of stiffness).
 - **Generalized 3D Formulation (Isotropic):** Relates 3D stresses and strains using Young's modulus (E) and Poisson's ratio (ν):

$$\epsilon_x = \frac{1}{E} [\sigma_x - \nu(\sigma_y + \sigma_z)] \quad (1)$$

$$\epsilon_y = \frac{1}{E} [\sigma_y - \nu(\sigma_x + \sigma_z)] \quad (2)$$

$$\epsilon_z = \frac{1}{E} [\sigma_z - \nu(\sigma_x + \sigma_y)] \quad (3)$$

- **Poisson's Ratio (ν):** The ratio of transverse strain to axial strain under tensile load:

$$\nu = -\frac{\epsilon_{\text{transverse}}}{\epsilon_{\text{axial}}} \quad (4)$$

2.2 Thermal Stress and Strain

When a material with Coefficient of Thermal Expansion (CTE, α) is heated by ΔT , its free thermal strain is:

$$\epsilon_{\text{th}} = \alpha \Delta T$$

If the material is clamped at both ends, prevented from expanding, it experiences a thermal stress:

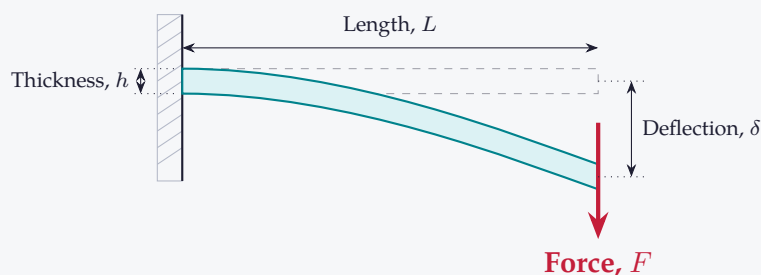
$$\sigma_{\text{th}} = -E\alpha\Delta T$$

This thermal stress is the primary driving mechanism in thermal actuators and a major source of thermal drift in capacitive sensors.

3. Cantilever Beam Deflection Derivation

Consider a rectangular cantilever beam of length L , width w , and thickness h clamped at $x = 0$, with a point force F applied downward at the free end $x = L$.

Cantilever Beam Deflection Schematic



The bending moment at any position x along the beam length is:

$$M(x) = -F(L - x)$$

According to Euler-Bernoulli beam theory:

$$EI \frac{d^2 y}{dx^2} = M(x) = -F(L - x)$$

where I is the rectangular moment of inertia:

$$I = \frac{wh^3}{12}$$

Integrate the differential equation once with respect to x :

$$EI \frac{dy}{dx} = -F \left(Lx - \frac{x^2}{2} \right) + C_1$$

Applying the boundary condition at the clamped end ($x = 0$): $\frac{dy}{dx} = 0 \Rightarrow C_1 = 0$. Integrate a second time:

$$EI y(x) = -F \left(L \frac{x^2}{2} - \frac{x^3}{6} \right) + C_2$$

Applying the boundary condition at $x = 0$: $y(0) = 0 \Rightarrow C_2 = 0$. Thus, the deflection curve of the beam is:

$$y(x) = -\frac{F}{6EI} (3Lx^2 - x^3)$$

The maximum deflection δ occurs at the free end $x = L$:

$$\delta = |y(L)| = \frac{FL^3}{3EI}$$

Using the spring relation $F = k\delta$, the equivalent spring constant k is:

$$k = \frac{3EI}{L^3}$$

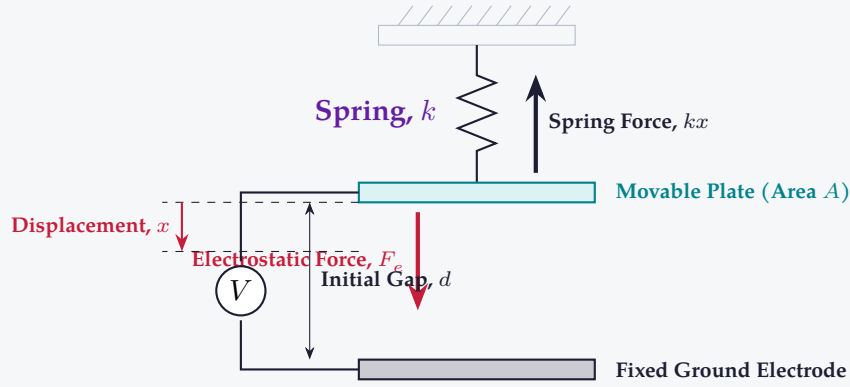
Substituting $I = \frac{wh^3}{12}$:

$$k = \frac{3E}{L^3} \left(\frac{wh^3}{12} \right) = \frac{Ewh^3}{4L^3}$$

4. Coupled Electromechanical Pull-in Model

An electrostatic parallel-plate actuator experiences pull-in instability, where electrostatic forces overwhelm mechanical spring restoring forces.

Parallel-Plate Spring-Actuator Model



The force balance equation of the system at equilibrium is:

$$F_s = F_e \implies kx = \frac{1}{2} \frac{\epsilon A V^2}{(d-x)^2}$$

where k is spring constant, d is initial gap, and x is displacement. Solve for V^2 :

$$V^2 = \frac{2kx(d-x)^2}{\epsilon A}$$

To find the limits of stable equilibrium, we find the maximum voltage that can be applied by taking the derivative with respect to x and setting it to zero:

$$\begin{aligned} \frac{d(V^2)}{dx} = 0 &\implies \frac{2k}{\epsilon A} \frac{d}{dx} [x(d-x)^2] = 0 \\ \frac{d}{dx} (xd^2 - 2dx^2 + x^3) &= d^2 - 4dx + 3x^2 = 0 \\ (d-3x)(d-x) &= 0 \end{aligned}$$

This yields two solutions: $x = d$ (unstable limit where gap is zero) and:

$$x_{pi} = \frac{d}{3}$$

Conclusion 1: The critical displacement at the pull-in point is exactly one-third of the initial gap. Substitute $x_{pi} = d/3$ back into the voltage equation to find the critical **Pull-in Voltage** (V_{pi}):

$$\begin{aligned} V_{pi}^2 &= \frac{2k \left(\frac{d}{3}\right) \left(d - \frac{d}{3}\right)^2}{\epsilon A} = \frac{2k \left(\frac{d}{3}\right) \left(\frac{2d}{3}\right)^2}{\epsilon A} = \frac{8kd^3}{27\epsilon A} \\ V_{pi} &= \sqrt{\frac{8kd^3}{27\epsilon A}} \end{aligned}$$

5. Finite Element Method (FEM) Overview

Analytical models are insufficient for complex structures. The Finite Element Method (FEM) provides numerical solutions by dividing a continuous model into finite sub-regions.

- **Discretization (Meshing):** Subdividing the structure into discrete shapes (elements) connected at node points.
- **Governing Equations:** Formulation of algebraic equations for each element based on differential equations (e.g., Navier-Cauchy equations for elasticity).
- **Coupled-Field Analysis:** Crucial for MEMS, it simulates interactions across multiple physical domains concurrently (e.g., electrostatic field solving updates structural deflection, which in turn alters the electrostatic mesh).

Quick Revision — Module IV

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Key Mechanical & Electrostatic Equations

Parameter	Core Equation	Key Variable Definition
Hooke's Law (1D)	$\sigma = E\epsilon$	Relates mechanical stress and strain
Poisson's Ratio	$\nu = -\epsilon_{\text{trans}}/\epsilon_{\text{axial}}$	Ratio of transverse to axial strain
Thermal Stress	$\sigma_{\text{th}} = -E\alpha\Delta T$	E : Young Modulus; α : CTE; ΔT : Temp difference
Rect. Inertia	$I = wh^3/12$	w : width; h : thickness of beam
Cantilever Spring Const.	$k = 3EI/L^3 = Ewh^3/4L^3$	L : Beam length; I : Moment of Inertia
Pull-in Voltage	$V_{\text{pi}} = \sqrt{\frac{8kd^3}{27\epsilon A}}$	k : Spring constant; d : gap; A : Plate area
Pull-in Displacement	$x_{\text{pi}} = d/3$	Deflection limit before electro-snapdown

One-Line Definitions

- **Bulk Micromachining:** A fabrication technique where substrate material is removed to define micro-structures.
- **HNA:** An isotropic silicon wet etchant consisting of Hydrofluoric acid, Nitric acid, and Acetic acid.
- **Anisotropic Etch:** A directional etch that exhibits different rates along different crystallographic planes (e.g., KOH).
- **Anodic Bonding:** A bonding method joining silicon to sodium-rich glass using heat (400°C) and high voltage.
- **Eutectic Bonding:** A bonding process using a gold-silicon interface heated to 363°C to form a liquid-alloy bond.
- **Hooke's Law:** Linear relationship between stress and strain within the elastic limit of a material.
- **Young's Modulus (E):** A measure of the elastic stiffness of a solid material; equal to stress divided by strain.
- **Pull-in Instability:** The electrostatic collapse of a suspended electrode when displacement exceeds one-third of the gap.
- **FEM:** A numerical modeling technique that solves complex engineering physics problems by discretization (meshing).

Frequently Asked / Viva Questions

Q1. Why does KOH etch the {111} plane of Silicon much slower than the {100} plane?

Because the {111} plane has the highest atomic packing density, and surface atoms on {111} expose only a single dangling bond (compared to two on {100}), making it chemically highly stable.

Q2. Calculate the sidewall slope angle of a KOH-etched pyramidal cavity on {100} silicon.

The angle between the {100} plane and the {111} plane is exactly 54.74° .

Q3. Compare direct bonding and anodic bonding in MEMS packaging.

Direct bonding links silicon-to-silicon at high temperatures ($> 1000^\circ\text{C}$). Anodic bonding links silicon-to-glass at lower temperatures ($300 - 450^\circ\text{C}$) using a high electrostatic field.

Q4. Explain the physical meaning of Poisson's ratio (ν).

It is the ratio of lateral contraction strain to longitudinal extension strain under tensile stress. For silicon, $\nu \approx 0.28$.

Q5. State Hooke's Law and its generalized formulation in 3D for isotropic materials.

Hooke's Law states that stress is directly proportional to strain: $\sigma = E\epsilon$. In 3D, taking into account transverse Poisson contraction, the strains are: $\epsilon_x = \frac{1}{E}[\sigma_x - \nu(\sigma_y + \sigma_z)]$ (symmetrically for ϵ_y and ϵ_z).

Q6. How is thermal stress generated in clamped micro-structures?

If a micro-beam is anchored at both ends, thermal expansion is restricted. Heating by ΔT induces compressive thermal stress $\sigma_{\text{th}} = -E\alpha\Delta T$.

Q7. State the spring constant equation of a rectangular cantilever beam and its dependency on parameters.

$k = \frac{Ewh^3}{4L^3}$. It is directly proportional to width (w) and the cube of thickness (h^3), and inversely proportional to the cube of length (L^3).

Q8. Derive why pull-in displacement occurs at exactly one-third of the initial gap ($d/3$).

By setting the derivative of voltage squared $\frac{d(V^2)}{dx} = 0$ for the force-balance equation $kx = \frac{\epsilon AV^2}{2(d-x)^2}$, we obtain the equation $d^2 - 4dx + 3x^2 = 0$, which factorizes to yield $x_{\text{pi}} = d/3$.

Q9. What happens to an electrostatic parallel-plate actuator when the voltage exceeds V_{pi} ?

The electrostatic force increases faster than the spring restoring force, causing the movable plate to collapse and snap down onto the fixed electrode (pull-in).

Q10. Why is coupled-field analysis necessary in FEM simulation of MEMS?

Frequently Asked / Viva Questions (contd.)

Because MEMS involves coupled physics (e.g. electromechanical). Electrostatic fields deform structures, which changes the electrostatic gap, altering the field. The solver must iteratively couple both solvers.

Q11. Detail the chemical mechanism of isotropic silicon etching in HNA.

Nitric acid (HNO_3) oxidizes Silicon to form SiO_2 , Hydrofluoric acid (HF) dissolves the SiO_2 , and Acetic acid (CH_3COOH) dilutes and buffers the reaction.